

Artificial Intelligence

AI2002

Rushda Muneer

Spring 2026

Monotonicity

- The definition of A* algorithms did not require that $g(n) = g^*(n)$.
- This means that **admissible heuristics may initially reach non-goal states along a suboptimal path**, as long as the algorithm eventually finds an optimal path to all states on the path to a goal.
- It is natural to ask **if there are heuristics that are “locally admissible,”** i.e., that **consistently find the minimal path to each state** they encounter in the search. This property is called monotonicity or consistency.
- When a state is discovered by using heuristic search, is there any guarantee that the same state won't be found later in the search at a cheaper cost (with a shorter path from the start state)

Monotonicity

- A heuristic function h is monotone if
 1. For all states n_i and n_j , where n_j is a descendant of n_i ,
 - $h(n_i) - h(n_j) \leq \text{cost}(n_i, n_j)$,
 - where $\text{cost}(n_i, n_j)$ is the actual cost (in number of moves) of going from state n_i to n_j .
 2. The heuristic evaluation of the goal state is zero, or $h(\text{Goal}) = 0$.
- The difference between the heuristic measure for a state and any one of its successors is bound by the actual cost of going between that state and its successor.

Monotonicity

- A simple argument can show that any monotonic heuristic is admissible.
- This argument considers any path in the space as a sequence of states $s_1, s_2, \dots, s_g$, where s_1 is the start state and s_g is the goal.
- For the sequence of moves in this arbitrarily selected path:
 - s_1 to s_2 $h(s_1) - h(s_2) \leq \text{cost}(s_1, s_2)$ by monotone property
 - s_2 to s_3 $h(s_2) - h(s_3) \leq \text{cost}(s_2, s_3)$ by monotone property
 - s_3 to s_4 $h(s_3) - h(s_4) \leq \text{cost}(s_3, s_4)$ by monotone property
 - $\cdot$ $\cdot$ $\cdot$ $\cdot$ $\cdot$ by monotone property
 - $\cdot$ $\cdot$ $\cdot$ $\cdot$ $\cdot$ by monotone property
 - s_{g-1} to s_g $h(s_{g-1}) - h(s_g) \leq \text{cost}(s_{g-1}, s_g)$ by monotone property
- Summing each column and using the monotone property of $h(s_g) = 0$:
- path s_1 to s_g $h(s_1) \leq \text{cost}(s_1, s_g)$
- This means that monotone heuristic h is A* and admissible

Example

Admissible?

$s \Rightarrow 4 \leq 10$

$a \Rightarrow 3 \leq 10$

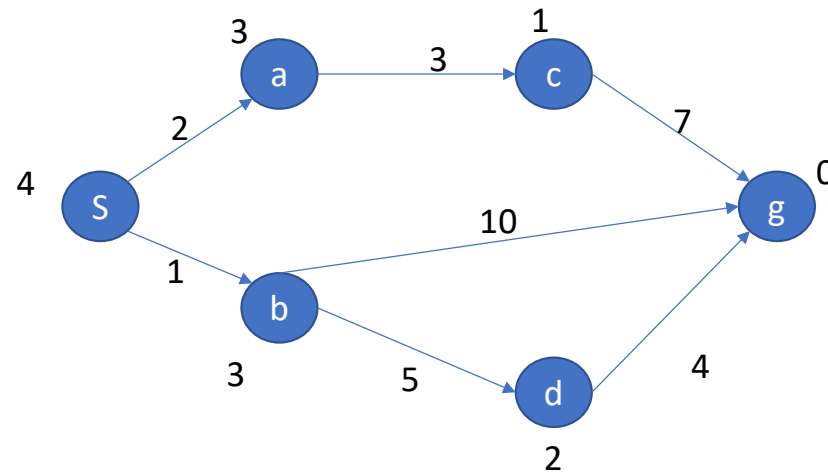
$b \Rightarrow 3 \leq 9$

$c \Rightarrow 1 \leq 7$

$d \Rightarrow 2 \leq 4$

$g \Rightarrow 0 \leq 0$

YES!



Monotone?

$s \Rightarrow a \quad 4 - 3 = 1 \leq 2$

$s \Rightarrow b \quad 4 - 3 = 1 \leq 1$

$a \Rightarrow c \quad 3 - 1 = 2 \leq 3$

$b \Rightarrow d \quad 3 - 2 = 1 \leq 5$

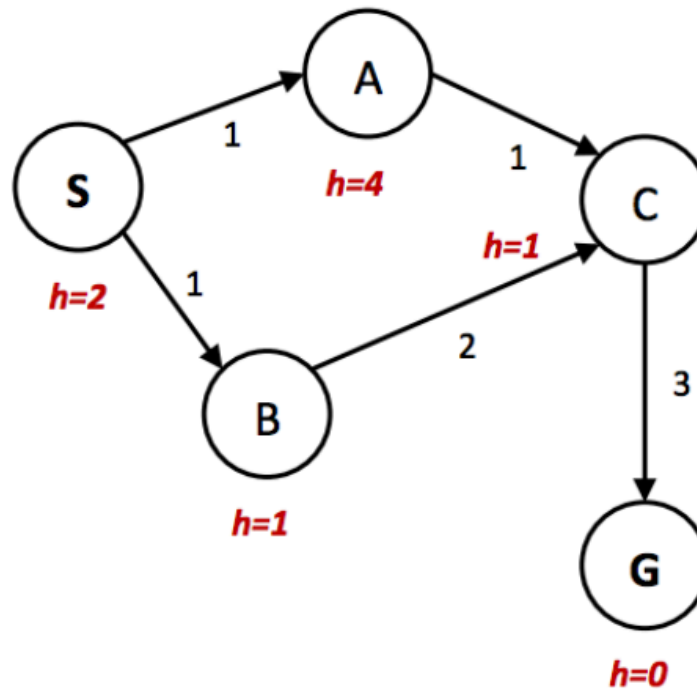
$b \Rightarrow g \quad 3 - 0 = 3 \leq 10$

$c \Rightarrow g \quad 1 - 0 = 1 \leq 7$

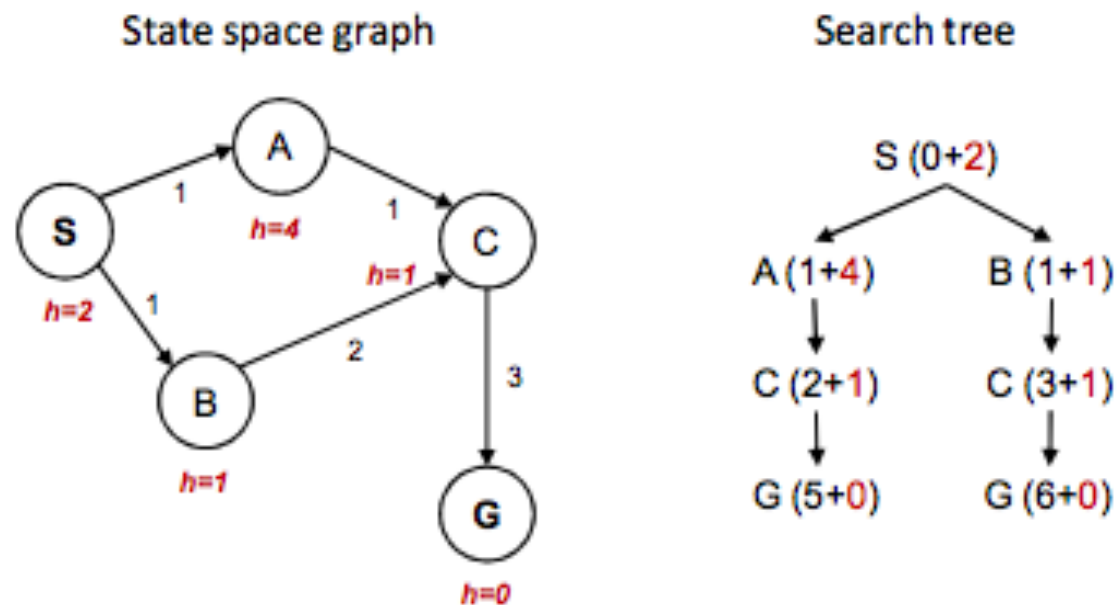
$d \Rightarrow g \quad 2 - 0 = 2 \leq 4$

YES

Activity: Admissible and Monotone heuristics

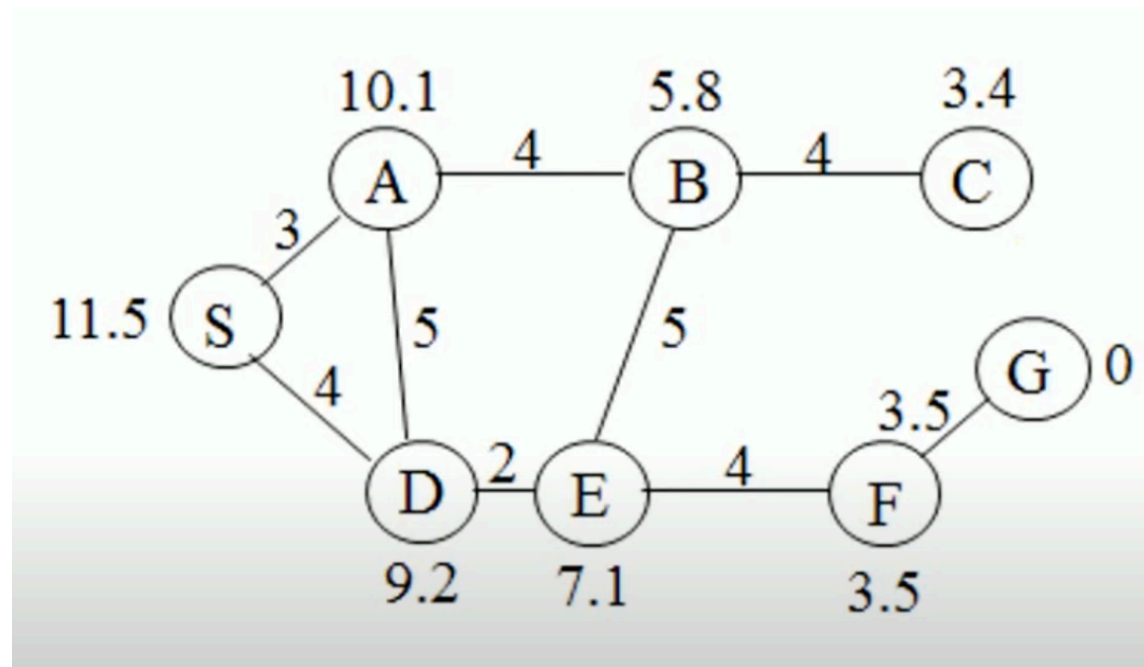


Activity: Admissible and Monotone heuristics



$h(A)$ makes the heuristics non-monotone

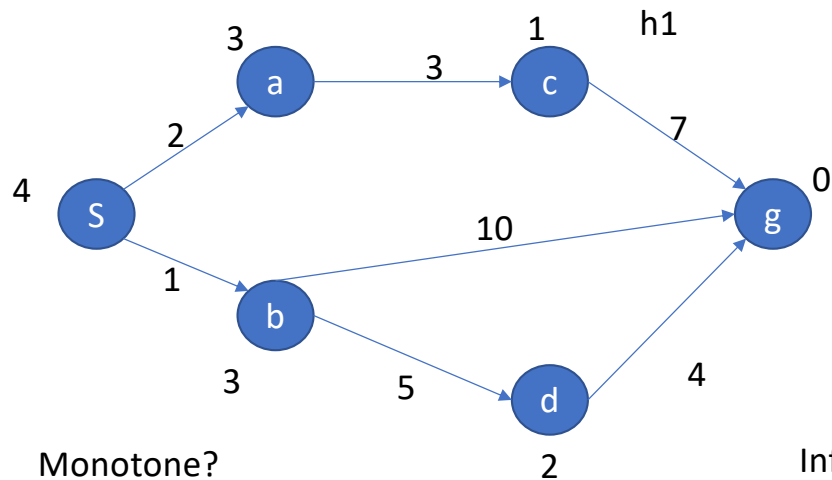
Activity Admissible and Monotone heuristics



Informedness

- For two A* heuristics h_1 and h_2 , if $h_1(n) \leq h_2(n)$, for all states n in the search space, heuristic h_2 is said to be **more informed** than h_1 .
- Informedness shows which heuristic is better than the other.
- For the 8-puzzle problem,
 - Breadth-first search is equivalent to the A* algorithm with heuristic h_1 such that $h_1(x) = 0$ for all states x . This is, trivially, less than h^* .
 - We have also shown that h_2 , the number of tiles out of place with respect to the goal state, is a lower bound for h^* .
 - In this case $h_1 \leq h_2 \leq h^*$.
- It follows that the “number of tiles out of place” heuristic is more informed than breadth-first search.
- Similarly, we can argue that the heuristic that calculates the sum of the direct distances by which all the tiles are out of place is again more informed than the calculation of the number of tiles that are out of place with respect to the goal state, and indeed this is the case.

Example



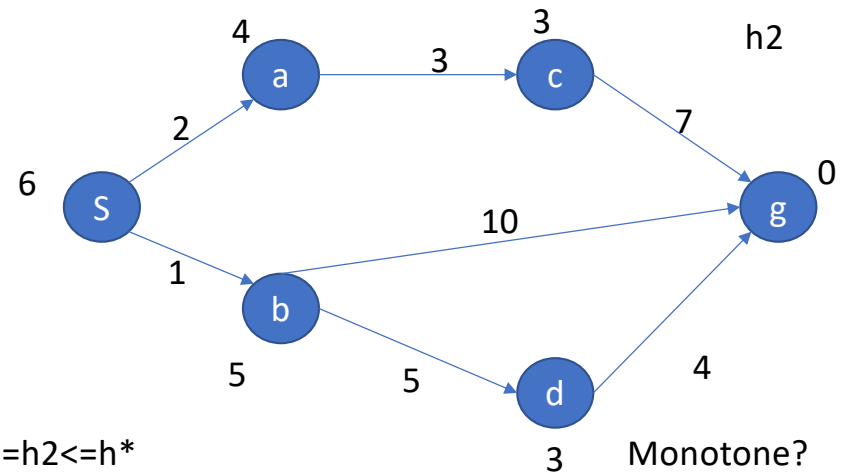
Monotone?

$s \Rightarrow a \quad 4 - 3 = 1 \leq 2$
 $s \Rightarrow b \quad 4 - 3 = 1 \leq 1$
 $a \Rightarrow c \quad 3 - 1 = 2 \leq 3$
 $b \Rightarrow d \quad 3 - 2 = 1 \leq 5$
 $b \Rightarrow g \quad 3 - 0 = 3 \leq 10$
 $c \Rightarrow g \quad 1 - 0 = 1 \leq 7$
 $d \Rightarrow g \quad 2 - 0 = 2 \leq 4$
 YES

Informedness $h1 \leq h2 \leq h^*$

$S \Rightarrow 4 \leq 6 \leq 10$
 $A \Rightarrow 3 \leq 4 \leq 10$
 $B \Rightarrow 3 \leq 5 \leq 9$
 $C \Rightarrow 1 \leq 3 \leq 7$
 $D \Rightarrow 2 \leq 3 \leq 4$

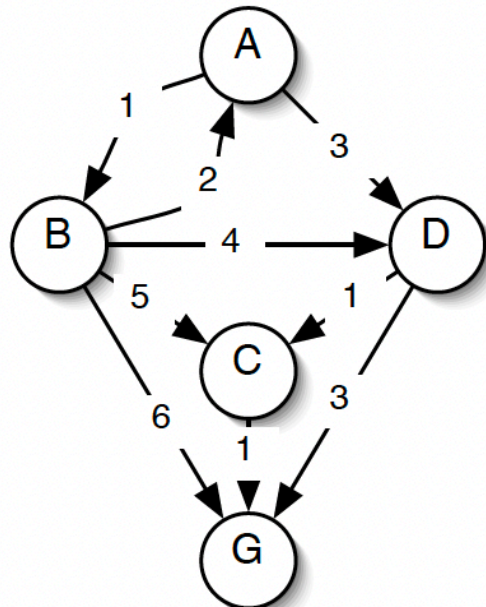
H2 are more informed than H1



Monotone?

$s \Rightarrow a \quad 6 - 4 = 2 \leq 2$
 $s \Rightarrow b \quad 6 - 5 = 1 \leq 1$
 $a \Rightarrow c \quad 4 - 3 = 1 \leq 3$
 $b \Rightarrow d \quad 5 - 3 = 2 \leq 5$
 $b \Rightarrow g \quad 5 - 0 = 5 \leq 10$
 $c \Rightarrow g \quad 3 - 0 = 3 \leq 7$
 $d \Rightarrow g \quad 3 - 0 = 3 \leq 4$
 YES

Activity: Admissibility, Monotonicity & Informedness?



	H1	H2
A	3	3
B	6	3
C	4	0
D	3	2

Readings

- G. F. Lugar – 4.3
- Modern approach – 3.5, 3.6